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SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 3

Design and Develop Model

COURSE NAME: Deep Learning
SLOT : D

COURSE CODE: MLA0405

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

Course Outcome Weightage: 10%
Assessment Tool Weightage: 30%

Duration: 90 minutes
Mark : 25

Code of Conduct:

I Yoga Madha A-192425058 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student

Name of the student: Yoga Madha A

Criteria	Article Selection & Problem Understanding (2)	Deep Learning Model & Methodology Analysis (2.5)	Results & Performance Evaluation (2.5)	Critical Analysis & Technical Evaluation (2)	Future Scope, Presentation & Documentation (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100 %
Q1						
Total						

QUESTIONS: 01

Total Marks:25

Article Review

Selected Research Article

Topic: DenseNet / Deep Learning / Plant Disease Classification

Selected Article:

“Robust Denoising and DenseNet Classification Framework for Plant Disease Detection”

The selected paper was published in **2024** and presents a DenseNet-based approach for citrus plant disease classification. The research is particularly relevant to the Unit IV topics because it directly applies **DenseNet architecture, image classification, feature extraction, data augmentation, and deep-learning-based denoising.** ([DOI](#))

1. Article Details

Title of the Research Article

Robust Denoising and DenseNet Classification Framework for Plant Disease Detection

Authors

- **Kevin Zhou**
- **Dimah Dera**

The authors are affiliated with the University of Texas Rio Grande Valley and Rochester Institute of Technology, respectively. ([DOI](#))

Conference

Proceedings of the 19th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications (VISIGRAPP 2024) – Volume 3: VISAPP

Year of Publication

2024

Pages

166–174

DOI

10.5220/0012390400003660 ([DOI](#))

[Official publication page](#)

Topic / Deep Learning Technique

The article addresses:

- DenseNet
- CNN-based image classification
- Plant disease detection
- Image denoising
- Data augmentation
- Robust deep learning
- Adversarial-noise resistance

2. Research Problem

Plant diseases can significantly reduce agricultural productivity. Early identification of diseases is therefore important for preventing crop losses.

Traditional plant disease identification methods generally depend on:

- Manual inspection
- Agricultural experts
- Laboratory testing
- Traditional image-processing techniques

These methods can be time-consuming, expensive, and difficult to apply on a large scale. The researchers identified another important problem: many deep-learning plant-disease models are trained mainly using **clean images**. However, images obtained in real agricultural environments can contain:

- Gaussian noise
- Salt-and-pepper noise
- Environmental disturbances
- Image corruption
- Adversarial noise

Such noise can reduce classification performance and potentially cause incorrect predictions. The paper therefore proposes combining a **DenseNet classification model with deep-learning-based denoising models** to improve robustness against noisy and adversarial inputs.

Limitations of Existing Approaches

Existing approaches can suffer from:

1. Dependence on clean training images.
2. Reduced performance when images contain noise.
3. Vulnerability to adversarial attacks.
4. Traditional filtering methods may distort image quality.
5. Manual disease identification requires expert knowledge.

3. Objectives of the Study

The main objectives of the research are:

1. To develop a **DenseNet-based model** for citrus plant disease classification.
2. To investigate the effect of **noise injection image augmentation**.
3. To evaluate different deep-learning denoising models.
4. To combine denoising networks with DenseNet.
5. To improve classification robustness against noisy images.
6. To investigate resistance against **FGSM and PGD adversarial attacks**.
7. To compare the performance of DenseNet with and without denoising components.

The researchers specifically evaluated three denoising networks:

- **CBDNet – Convolution Blind Denoising Network**
- **RIDNet – Real Image Denoising Network**
- **REDNet – Residual Encoder-Decoder Network**

RIDNet was found to be particularly effective when combined with DenseNet for adversarial-noise robustness. ([ScitePress](#))

4. Dataset Description

The research used a citrus plant image dataset assembled from multiple sources.

Dataset Sources

The researchers used:

- Manually collected citrus plant images.
- **PlantVillage dataset**
- **Colombian citrus fruits dataset**

Dataset Size

The total dataset contains:

17,306 citrus plant images.

Data Type

The dataset consists of **images of citrus leaves and fruits.**

Six Classes

The images were categorized into six classes:

1. Black Spot
2. Canker
3. Citrus Greening Leaves
4. Citrus Greening Fruits
5. Healthy Fruits
6. Healthy Leaves

Dataset Division

The dataset was divided into:

Dataset	Percentage
Training	66%
Validation	16.5%
Testing	17.5%

This division was used for training, validation, and final evaluation.

Data Augmentation

The researchers used image-processing and augmentation operations such as:

- Cropping
- Zooming
- Noise injection
- Gaussian noise
- Salt-and-pepper noise

This helped the model learn to handle different image conditions.

5. Proposed Methodology

The proposed system consists of two main stages:

Stage 1 – Image Denoising

The researchers evaluated three deep-learning denoising models:

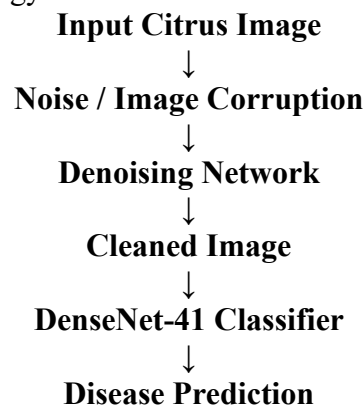
Input Image → CBDNet / RIDNet / REDNet → Denoised Image

Stage 2 – Disease Classification

The denoised image is then passed to the DenseNet classifier.

Denoised Image → DenseNet-41 → Disease Class

Therefore, the overall methodology is:



The researchers also trained DenseNet under two conditions:

1. Without noise-injection augmentation.
2. With noise-injection augmentation.

This allowed them to determine how noise augmentation affects robustness.

6. Deep Learning Technique Used

DenseNet

The primary deep-learning architecture is **DenseNet**.

DenseNet is a convolutional neural network in which layers are densely connected. Instead of passing information only from the immediately previous layer, later layers can receive feature information from earlier layers.

For example:

Layer 1 → Layer 2 → Layer 3 → Layer 4

In a DenseNet:

Layer 1 → Layer 2, Layer 3, Layer 4

Layer 2 → Layer 3, Layer 4

Layer 3 → Layer 4

This encourages **feature reuse**.

DenseNet-41

The proposed classification architecture is called **DenseNet-41**.

It contains:

- Initial **7×7 convolution**
- Three dense blocks
- Three transition layers
- Dense blocks containing **4, 6, and 8 convolution pairs**
- One fully connected classification layer

The paper describes the model as having **41 convolution layers and one fully connected layer**.

Dense Block

Each dense block uses pairs of:

- **1×1 convolution**
- **3×3 convolution**
- ReLU activation
- Batch normalization

Transition Layer

Each transition layer contains:

- 1×1 convolution
- ReLU
- Batch normalization
- 2×2 average pooling

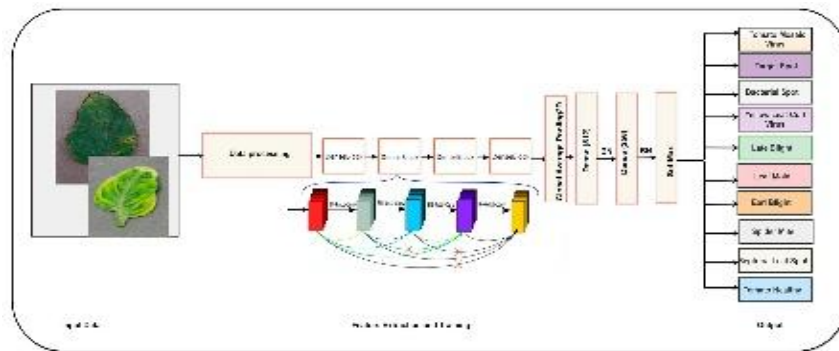
Why DenseNet?

DenseNet is appropriate because:

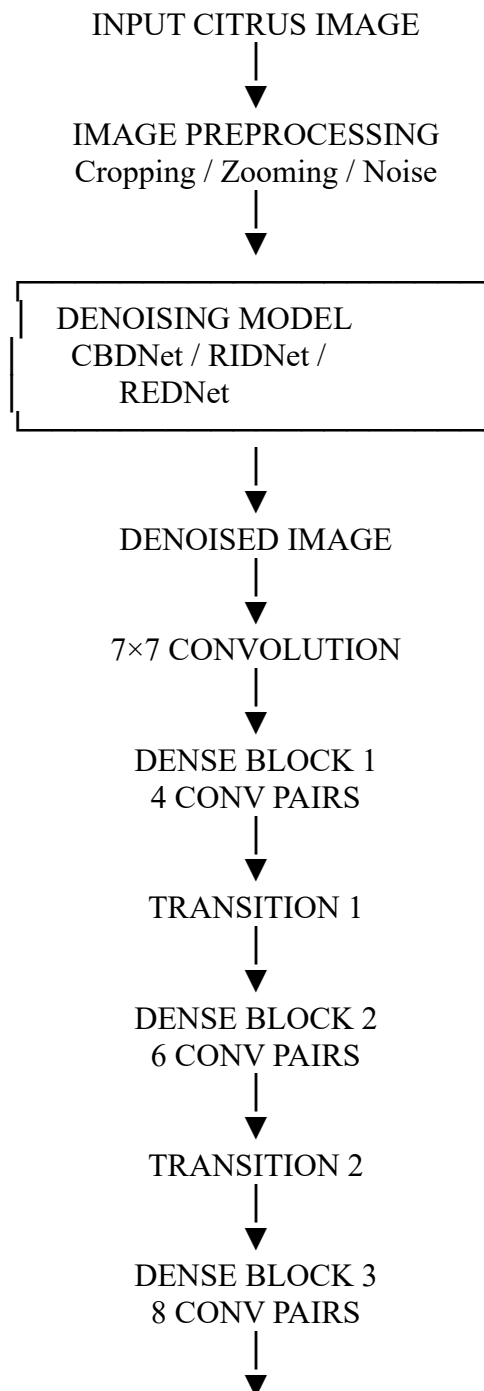
- Features can be reused.
- Gradient flow is improved.
- Earlier information can reach deeper layers.
- It can efficiently learn visual disease patterns.
- It is effective for image classification.

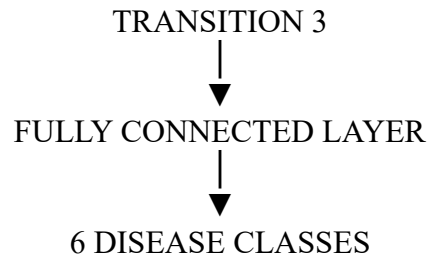
7. Model Architecture

Simplified Architecture



Proposed Architecture





Role of Major Components

Component	Function
Input	Receives citrus leaf/fruit image
Preprocessing	Improves image and introduces/handles noise conditions
Denoising network	Removes unwanted image noise
Convolution layers	Extract visual features
Dense blocks	Reuse previously learned features
Transition layers	Reduce feature-map size
Fully connected layer	Performs final classification
Output	Predicts one of six citrus categories

8. Performance Evaluation

The researchers evaluated the classification system using:

- **Accuracy**
- **Precision**
- **Recall**
- **F1-score**
- **Loss**
- **PSNR**
- **SSIM**

For adversarial robustness, the researchers evaluated the model against:

- **FGSM – Fast Gradient Sign Method**
- **PGD – Projected Gradient Descent**

Classification Results

For the DenseNet-41 model with noise augmentation, the reported overall training accuracy was approximately **96.90%** for the ground-truth image condition, while Gaussian and salt-and-pepper noise conditions also produced approximately **96.77% and 96.94%** accuracy respectively. ([ScitePress](#))

The class-level results show that some classes achieved very high precision, recall, and F1-score.

For example, with noise augmentation:

Class	Precision	Recall	F1-score
Black Spot	0.79	0.95	0.86
Canker	0.87	0.48	0.62
Greening Fruit	0.97	0.94	0.96
Healthy Fruit	1.00	1.00	1.00
Greening Leaves	0.99	1.00	0.99
Healthy Leaves	0.97	1.00	0.98

9. Results and Discussion

The research demonstrated that the proposed DenseNet-based classification system can effectively classify citrus diseases.

Major Findings

1. DenseNet-41 achieved high classification accuracy.
2. Noise augmentation improved robustness.
3. Denoising networks improved resistance to corrupted inputs.
4. RIDNet was particularly effective against adversarial noise.
5. The denoising models produced improved image quality.
6. The combined system was more robust than using the classification model alone.

The researchers found that denoising models did **not produce a large improvement for ordinary Gaussian and salt-and-pepper noise**, because DenseNet-41 was already relatively resistant to those noise types. However, the denoising models were more valuable against **adversarial noise**. ([ScitePress](#))

Best Performing Denoising Model

RID-DNet, which combines RIDNet with DenseNet-41, was identified as the best-performing denoising framework in the adversarial-noise experiments.

For PGD attacks, the stabilized accuracy of RID-DNet was reported between approximately **89.71% and 92.01%**, while for FGSM attacks it was approximately **92.88% to 98.83%**, depending on noise severity. ([ScitePress](#))

10. Advantages of the Proposed Model

1. High Classification Performance

DenseNet-41 provides strong performance in classifying different citrus diseases.

2. Feature Reuse

Dense connections allow earlier features to be reused by later layers. This improves information flow and feature propagation.

3. Noise Robustness

The proposed framework can handle:

- Gaussian noise
- Salt-and-pepper noise
- FGSM adversarial noise
- PGD adversarial noise

4. Improved Image Quality

The denoising networks can improve the quality of corrupted images before classification.

5. Better Real-World Applicability

The approach addresses a practical limitation of models trained only on clean laboratory images.

6. Adversarial Defense

Combining denoising networks with DenseNet provides additional protection against adversarial perturbations.

11. Limitations of the Study

Although the research provides promising results, several limitations can be identified.

1. Dataset Limitations

The dataset contains 17,306 images, but the classes are not equally distributed. For example, the number of healthy fruit and healthy leaf samples is considerably larger than some disease categories.

2. Computational Complexity

The complete framework uses both a denoising network and a classification network. This can increase:

- Memory requirements
- Processing time
- Computational cost

3. Multiple Models

Using CBDNet, RIDNet, REDNet, and DenseNet increases system complexity compared with a single CNN classifier.

4. Limited Crop Scope

The research focuses specifically on **citrus plant diseases**. Therefore, the results may not automatically generalize to:

- Apple diseases
- Tomato diseases
- Potato diseases
- Rice diseases
- Other crops

5. Real-World Generalization

Although the study considers noisy and adversarial conditions, additional testing using large-scale images captured under diverse real agricultural environments would strengthen the conclusions.

6. Class Imbalance

Some classes have significantly more samples than others, which may influence classification performance.

12. Critical Analysis

Is DenseNet appropriate?

Yes.

DenseNet is appropriate because plant disease classification requires extraction of different visual characteristics such as:

- Leaf spots
- Color changes
- Texture
- Shape
- Disease patterns

Dense connections allow these features to be reused across deeper layers.

Is the dataset adequate?

The dataset of **17,306 images** provides a reasonable experimental base. However, greater diversity across geographical locations, lighting conditions, crop varieties, and field environments would improve generalization.

Are the evaluation metrics appropriate?

Yes. Accuracy alone is not sufficient for disease classification, particularly when classes are imbalanced.

The use of:

- Precision
- Recall
- F1-score
- Accuracy
- Loss

provides a more comprehensive evaluation.

The use of **PSNR and SSIM** is also appropriate for evaluating image denoising quality.

Are the results convincing?

The results are promising because the authors compare:

- DenseNet alone
- DenseNet with noise augmentation
- DenseNet combined with denoising networks
- Different noise types
- Different adversarial attacks

This provides useful evidence for robustness.

However, additional testing on completely independent real-world datasets would make the conclusions stronger.

Could another Unit IV technique provide better performance?

Possibly.

For example:

- **Transfer Learning** could be useful when training data is limited.
- **EfficientNet** could provide better computational efficiency.
- **Vision Transformers** could capture broader image relationships.
- **Attention mechanisms** could help focus on disease regions.

However, DenseNet remains a strong choice because feature reuse is particularly useful for visual classification.

13. Future Scope

The following improvements can be considered:

1. Larger and More Diverse Dataset

Collect images from:

- Different farms
- Different countries
- Different lighting conditions
- Different citrus varieties
- Different disease stages

2. More Plant Species

The system can be extended to:

- Tomato
- Potato
- Apple
- Grape
- Rice
- Wheat
- Maize

3. Mobile Application

The trained model could be deployed in a smartphone application where farmers capture a leaf image and immediately receive a disease prediction.

4. Lightweight Architecture

A lighter model such as MobileNet or EfficientNet could be investigated for real-time agricultural deployment.

5. Explainable AI

Techniques such as **Grad-CAM** could be incorporated to show the region of the leaf that influenced the prediction.

6. Advanced Attention Mechanisms

Attention mechanisms could help the model focus specifically on disease-infected regions.

7. Real-Time Field Testing

Future research should test the model on images captured directly from agricultural fields rather than relying mainly on existing datasets.

8. Improved Adversarial Defense

More advanced adversarial-defense mechanisms could be investigated to improve robustness against stronger attacks.

14. Conclusion

The reviewed research article, “**Robust Denoising and DenseNet Classification Framework for Plant Disease Detection**,” presents a robust deep-learning solution for citrus plant disease classification.

The proposed approach combines **DenseNet-41** with deep-learning-based denoising networks such as **CBDNet, RIDNet, and REDNet**. The system was evaluated using **17,306 citrus images** belonging to six categories. The research demonstrated that DenseNet can achieve high classification performance and that combining denoising networks with DenseNet improves robustness, particularly against adversarial noise.

Overall, the article is highly relevant to the **DenseNet and CNN topics** because it demonstrates the practical application of dense connections, feature reuse, image classification, data augmentation, and robust deep learning.

From a critical perspective, the proposed approach is technically appropriate and experimentally useful, but future work should focus on larger and more diverse real-world datasets, computationally efficient models, explainable AI, and real-time field deployment.

15. References

Primary Research Article

1. Zhou, K., & Dera, D. (2024). **Robust denoising and DenseNet classification framework for plant disease detection**. *Proceedings of the 19th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications (VISAPP)*, 166–174. <https://doi.org/10.5220/0012390400003660> (DOI)

Additional References

1. Huang, G., Liu, Z., Van Der Maaten, L., & Weinberger, K. Q. (2017). **Densely connected convolutional networks**. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 4700–4708.
2. Guo, S., Yan, Z., Zhang, K., Zuo, W., & Zhang, L. (2019). **Toward convolutional blind denoising of real photographs**. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
3. Mao, X., Shen, C., & Yang, Y.-B. (2016). **Image restoration using very deep convolutional encoder-decoder networks with symmetric skip connections**. *Advances in Neural Information Processing Systems*, 29.
4. Anwar, S., & Barnes, N. (2020). **Real image denoising with feature attention**. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*.
5. Goodfellow, I. J., Shlens, J., & Szegedy, C. (2015). **Explaining and harnessing adversarial examples**. *International Conference on Learning Representations (ICLR)*.
6. Madry, A., Makelov, A., Schmidt, L., Tsipras, D., & Vladu, A. (2018). **Towards deep learning models resistant to adversarial attacks**. *International Conference on Learning Representations (ICLR)*.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Article Selection & Problem Understanding	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Deep Learning Model & Methodology Analysis	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Results & Performance Evaluation	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Critical Analysis & Technical Evaluation	20%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Future Scope, Presentation & Documentation	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis